MIL-L-48664A (AR)
10 August 1988
SUPERSEDING
MIL-L-48664 (AR)
14 August 1986

MILITARY SPECIFICATION

LOOSE PROJECTILE RESTRAINT SYSTEM (LPRS), 155mm

This specification is approved for use within the US Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the material, requirements, quality assurance provisions and packaging of plastic divider racks used for restraining loose, unpalletized 155mm projectiles during on vehicle storage and transport (see 6.1). The rack consists of short and long plastic dividers that when assembled will restrain any conventional projectiles of the 155mm family (see 6.2). The loaded racks are restrained in a tactical transport vehicle by means of a cargo tiedown strap(s) applied in an approved manner. The combination is referred to as Loose Projectile Restraint System (LPRS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-595 -Color

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attritubes
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-1235	-Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other government documents, drawings and publications. The following other Government documents, drawings and publications form a part of the specification to the extent specified herein: Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

HANDBOOKS

MILITARY

MIL-HDBK-728/6	-Military Handbook, Ultrasonic Testing

TECHNICAL MANUALS

MILITARY

TM 9-2590-210-10	-Operators Manual: Loose Projectile Restraint System (LPRS) for use With Field Artillery Companion Vehicles

DRAWINGS

ARMY MATERIEL COMMAND

AC200000400	-Combination of Adopted Items, Loose Projectile Restraint System, 155mm
AC200000401	-Rack, Ammunition, 15 Round Assy, Loose Projectile Restraint System, 155mm
AC200000402	-Divider, Short, Loose Projectile Restraint System, 155mm

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AC200000403	-Divider, Long, Loose Projectile Restraint System, 155mm
AC200000404	-Packaging, Loose Projectile Restraint System, 155mm
AC200000405	-Test, Ease of Disassembly, Loose Projectile Restraint System
AC200000408	-Material for Divider, Ammunition, Loose Projectile Restraint System
9215220	-Projectile 155mm, M483A1 DPICM, LAP Assy
9216352	-Projectile 155mm, M107 HE, LAP Assy
9235999	-Projectile 155mm, M549A1 HE RAP, LAP Assy

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacture of the Short and Long Dividers, Ammunition, shall be as specified on Dwg AC200000408. Certification of compliance shall be provided by the contractor (see 4.1.2.1).

3.2 Dividers. Parts and assemblies shall comply with all requirements specified on the applicable drawings and with all requirements specified in the applicable specifications.

3.2.1 Flatness. When new and in a relaxed position laid flat on a side, the sides of a divider shall be straight within 1/8 inch from a 24 inch straight edge placed at any location other than over the stacking lugs when inspected as specified in 4.5.6.

3.2.2 Weld cones. Weld cones shall be homogeneously welded together at the plane of the mold parting line to provide maximum strength to the divider as intended by design when inspected as specified in 4.5.3.

3.2.3 Drainage holes. Drainage will be provided for any moisture accumulated within the divider interior when the dividers are assembled on either edge.

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3.2.4 Color. The color of the divider shall be olive drab matching any of the following colors of Federal Standard 595; -34052, -34079, -34086, -34087, -34096, or -34102. To retard sun degradation, ultraviolet stabilizer shall be added into the color concentrate. Certification shall be provided by the contractor (see 4.1.2.2) to assure that the ultraviolet stabilizer added into the color concentrate consists of 0.3 percent Tinuvin 770 and 0.05 percent Igranox 1076.

3.2.5 Stacking lugs. Short dividers or long dividers when laid flat and stacked, will nest on their stacking lugs without space between dividers. This nesting is required in either flat position, i.e., open slots pointing in the same direction or alternating directions. The vertical sides of the stack shall align within ± 1/8 inch. Long dividers and short dividers are not intermixed for this requirement (see 4.5.6).

3.3 Performance test. The rack, ammunition, 9, 15, and 25 round assemblies shall be subjected to the test specified in 4.5.1. Only the 15 round assembly shall be subjected to the test specified in 4.5.4. The following defects shall not be permitted:

a. Failure of the rack by deforming, splitting or breaking which allows the dividers to disengage and spill the projectiles during the performance test.

b. Separating of the parting line seam more than 2.5 inches along the top edge of the divider in contact with the projectiles. This failure shall not result in a collapse of the divider walls below the next structural stiffener; either handle or weld cones.

c. Any breaking or cracking of the rack other than at the parting line seam (see 3.3b) during the performance tests. Punctures, abrasions and wear through caused by the restraint straps metal parts or projectile grommets are not counted as a failure, providing such wear does not lead to functional failure of the dividers or rack assembly.

d. Excessive bending of the short sections at the corners of an assembled rack loaded with projectiles and restrained to a vehicle side wall by a cargo strap on three sides of the rack per TM 9-2590-21070. Excessive bending is deformation which allows the restraint strap(s) to disengage from the strap slots allowing the rack assembly to shift or upset during the performance test (see 4.5.1.2).

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e. Deformation of a divider which interfaces with ease of disassembly (see 4.5.5).

f. Failure of disassembled dividers to be reassembled into a load bearing rack for successful continuation of the performance test.

3.4 National Stock Number (NSN). The divider design shall include a thirteen digit (plus three dashes) National Stock Number molded in a recessed manner in each divider in 3/8 inch characters by 1/16 deep.

3.5 Ease of assembly and disassembly. The divider slots shall be so constructed to permit ease of assembly and disassembly yet the assembled divider rack must be rigid enough to maintain its original configuration while loaded. The loaded rack configuration is evaluated by the performance test (see 4.5.1) and the extreme operating temperature tests (see 4.5.4). Ease of disassembly is tested as specified in 4.5.5.

3.6 Extreme operating temperatures. The rack, ammunition must restrain 155mm projectiles at temperatures from minus 60 degrees fahrenheit to plus 145 degrees fahrenheit (see 4.5.4), as evidenced by meeting the performance test requirements of 3.3.

3.7 First article inspection. This specification makes technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. The dividers shall be regular, relatively smooth and free from wrinkles and rough spots to the extent that minor irregularities shall not interfere with successful testing under 4.5.5. Critical areas are within and adjacent to the assembly slots. The dividers shall be free from cracks, pinholes and any other defects that might affect serviceability and durability. Flashing at the parting lines shall not exceed 1/16 inch height on external perimeters and 1/32 inch height at slots and handles. Clean up shall include breaking of sharp corners. All dividers shall be free of chips, dirt, grease and foreign material. If cleaning agents are used, the method shall not be injurious to the divider material nor shall any of the dividers be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certification.

4.1.2.1 Certification of material (see 3.1). A certification of conformance shall be submitted by the material supplier/contractor for each lot utilized confirming the latest nominal physical properties and two of the basic material parameters which dictate the physical and chemical characteristics of the hexane copolymer material as listed on Dwg. AC200000408. Specifically the two basic material parameters are: density and melt flow. The nominal physical properties are based on typical values obtained from periodic testing of random lots of the same product. The basic and nominal physical properties shall be found to be compatible with the parameters as listed in Table II, Dwg. AC200000408. See this drawing for form of certification to be submitted as specified in the contract.

4.1.2.2 Certification of ultraviolet stabilizer. A certificate of conformance shall be submitted by the contractor, certifying that the ultraviolet stabilizer, added to the color concentrate, meets the requirements of 3.2.4. The method of certification shall be as specified in the contract.

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4.2 Classification of inspections. The following types of inspection shall be conducted on these items:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 72 dividers, ammunition, short, AMC Dwg. -AC200000402 from each mold and 56 dividers, ammunition, long, AMC Dwg. -AC200000403 from each mold which have been produced by the contractor using the same production processes, procedures, material and equipment as will be used in fulfilling the contract. The sample shall be selected proportionally from each cavity in the mold. The components shall be identified by mold number and the cavity number in the mold, if more than one mold per component is used.

4.3.1.1 Percent regrind for first article submission. To insure that production processes, procedures, and material are the same for First Article Submission as in production, the contractor will first determine the percent regrind which will be used in production. This percentage shall not exceed 45 percent. Prior to saving any sample for First Article Submission, the material mix to the extruder will be run through 5 iterations to insure the first sample saved contains 2 percent or less of material that has seen five or more melt cycles. For purposes of establishing the regrind mix through five iterations, flashing and parts may be used for regrind material. The following discussion to clarify the fifth iteration is based on an assumed 45 percent regrind: During the initial machine set-up, the parts and flashing produced will consist of 100 percent virgin material (designated as A_0). Once regrind is generated from iteration A_0 and is introduced into the melt cycle, the parts and flashing will consist of 55 percent virgin and 45 percent "one time through" regrind (one time through is designated as A_1). When the flashing from these parts is again introduced into the melt cycle, the resulting material composition of the parts and flashing consists of 55 percent virgin and 45 percent "once and twice through" regrind (twice through is A_2, three times is A_3, etc.). The second and subsequent iteration mixes are:

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(2) $.55A_0 + .45(.55A_1 + .45A_2) = .55A_0 + (.2475A_1 + .2025A_2)$

(3) $.55A_0 + .45(.55A_1 + .2475A_2 + .2025A_3) = .55A_0 + (.2475A_1 + .1114A_2 + .0911A_3)$

(4) $.55A_0 + .45(.55A_1 + .2475A_2 + .1114A_3 + .0911A_4) = .55A_0 + (.2475A_1 + .1114A_2 + .0501A_3 + .010A_4)$

(5) $.55A_0 + .45(.55A_1 + .2475A_2 + .1114A_3 + .0501A_4 + .0410A_5) = .55A_0 + (.2475A_1 + .1114A_2 + .0501A_3 + .0225A_4 + .0184A_5)$

4.3.2 Inspections to be performed. Samples may be subjected by the Government to any or all of the examinations and tests specified in Table I and any or all requirements of the applicable drawings.

4.3.3 Rejection. If any divider fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the sample to comply with any of the stated requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-L-48664A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Divider, Ammunition, Short, LPRS 155mm Divider, Ammunnition, Long, LPRS 155mm Rack Ammunition, 15 round assy, LPRS, 155mm				AC200000401, 402, 403
					NEXT HIGHER ASSEMBLY AC200000400
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Divider, Ammunition, Short See Applicable Drawing: AC200000402				
	(1) Examination for Defects.	28	-	3.2	4.4.2.1
	(2) Destructive test for conformation of weld cone weldment.	4	-	3.2.2	4.5.3/SMTE
	Divider, Ammunition, Long See Applicable Drawing: AC200000403				
	(1) Examination for Defects.	20	-	3.2	4.4.2.2/SMTE
	(2) Destructive test for conformation of weld cone	4	-	3.2.2	4.5.3/SMTE
	Rack, Ammunition, 15 Round Assembly See Applicable Drawing: AC200000401				
	(1) Ease of Assembly and disassembly	6Racks	-	3.5	4.5.5 Spring Scale
	(2) Performance Test		-	3.3	4.5.1
	a. 9 round rack configuration	2Racks			
	b. 15 round rack configuration	2Racks			
	c. 25 round rack configuration	1Rack			
	(3) Extreme Operating Temperature Test	4Racks		3.3	4.5.4

NOTES:
Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.4.2 Examination. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plan and procedures for major and minor defects shall be in accordance with MIL-STD-105. Inspection Level S-4. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER AC200000402
4.4.2.1	Divider, Ammunition, Short, LPRS, 155mm				NEXT HIGHER ASSEMBLY AC200000400
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Fold lines		0.40%	3.2	4.5.7/Visual
102	Small width of open slot (.625)		0.40%	3.2	SMTE (see 6.4)
103	Large width of open slot (1.190)		0.40%	3.2	SMTE
104	Thickness of open slot (.575		0.40%	3.2	SMTE
105	Width of closed slot (.625)		0.40%	3.2	SMTE
106	Thickness of closed slot (.575)		0.40%	3.2	SMTE
107	Thickness of panel both sides of slot, closed edge to center line (1.130)		0.40%	3.2	SMTE
108	Distance center to center of slots (8-1/8)		0.40%	3.2	SMTE
109	Location and dimensions of stacking lugs		0.40%	3.2.5	4.5.6/SMTE
110	Flatness		0.40%	3.2.1	4.5.6/SMTE
111	Wall thickness		0.40%	3.2	4.5.2/SMTE
Minor					
201	Overall length of panel		0.65%	3.2	SMTE
202	Overall height of panel		0.65%	3.2	SMTE
203	Color		0.65%	3.2.4	Visual
204	Drainage holes		0.65%	3.2.3	Visual
205	Legibility of NSN		0.65%	3.4	Visual
206	Workmanship		0.65%	3.8	Visual/measure

NOTES:

Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Divider, Ammunition, Long, LPRS, 155mm	1 OF 1	AC200000403
			NEXT HIGHER ASSEMBLY: AC200000400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Fold lines		0.40%	3.2	4.5.7/Visual
102	Small width of open slot (.625)		0.40%	3.2	SMTE (See 6.4)
103	Large width of open slot (1.190)		0.40%	3.2	SMTE
104	Thickness of open slot (.575)		0.40%	3.2	SMTE
105	Width of closed slot (.625)		0.40%	3.2	SMTE
106	Thickness of closed slot (.575)		0.40%	3.2	SMTE
107	Thickness of panel both sides of slot, closed edge to center line. (1.130)		0.40%	3.2	SME
108	Distance center to center of slots (8-1/8)		0.40%	3.2	SMTE
109	Location and dimensions of stacking lugs		0.40%	3.2.5	4.5.6/SMTE
110	Flatness		0.40%	3.2.1	4.5.6/SMTE
111	Wall thickness		0.40%	3.2	4.5.2/SMTE
Minor					
201	Overall length of panel		0.65%	3.2	SMTE
202	Overall height of panel		0.65%	3.2	SMTE
203	Color		0.65%	3.2.4	Visual
204	Drainage holes		0.65%	3.2.3	Visual
205	Legibility of NSN		0.65%	3.4	Visual
206	Workmanship		0.65%	3.8	Visual/measure

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions is performed.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-48664A (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.3	Rack, Ammunition, 15 Round Assembly, LPRS, 155mm		1	1	AC200000401
					NEXT HIGHER ASSEMBLY: AC200000400
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major 101	Ease of assembly and disassembly		0.40%	3.5	4.5.5/SMTE (See 6.4)

NOTES:
Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before the above test is performed.

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4.4.3 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all test and examinations specified in 4.4.2 and 4.5. The contractor is responsible for assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.5).

4.5 Methods of inspection.

4.5.1 Performance test (first article only). The performance test is to measure the physical strength of racks assembled from the divider samples. The test will be performed at the U.S. Army Defense Ammunition Center and School Test Facility located at Savanna, Illinois. The test vehicle will be a 5 Ton Tactical Cargo Truck. The various test loads will be arranged within the assembled rack configurations on the test vehicle and restrained by previously approved tie down procedures (see 6.6).

4.5.1.1 Test load. The test load will consist of a number of inert loaded 155mm projectiles as designated in 4.5.1.3.

PROJECTILE MODEL	DRAWING	PROJECTILE WEIGHT, pounds	LOCATION of CG, in. from base
M107	9216352	95.0	9.45
M483	9215220	103.3	13.20
M549	9235999	95.2	12.82

4.5.1.2 Road hazard course. The test course consist of two groups of alternating railroad ties spaced 8 to 10 feet apart within the group and 75 feet between the two groups. This course induces a 1 1/2 g vertical force and a 2 g lateral force to the test load. The initial tightening of the cargo restraint strap approximates 500 pounds tension and resultant initial force on the corners of the rack assembly.

4.5.1.3 Test configurations.

TEST	RACK CONFIGURATION	RACK LOCATION	TEST LOAD* EACH RACK	TEST DIVIDER SHORT LONG
1	2 ea 9 round	Each side of vehicle over wheels	3 ea M483 3 ea M549 3 ea M107	16

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2	2 ea 15 round	Each side of vehicle over wheels, long dividers parallel to side of vehicle	6 ea M483 3 ea M549 6 ea M107		12	8
3	1 ea 25 round	Against left side of vehicle over wheels	10 ea M483 5 ea M549 10 ea M107			12
				Total Samples	28	20

* Similar rounds are placed in adjacent locations laterally across the width of the vehicle.

4.5.1.4 Test. Each test will consist of six passes over the road hazard course. At the end of each pass, the dividers will be visually examined without dismantling the load. See 3.3 for acceptance criteria.

4.5.2 Wall thickness. Wall thickness shall be measured by ultrasonic test methods (MIL-HDBK-728/6 applies). The preferred method shall be contact testing with a suitable thin liquid film used as a couplant. The calibration reference standard will be a flat material of known thickness and matched acoustic properties. Wall thickness readings shall be taken at locations shown on sheet 2, AMC Dwg. AC200000402 and sheet 2, AMC Dwg. AC200000403. See notes 4 and 5 on the above drawings.

4.5.2.1 For each long divider inspected (see Dwg. AC200000403). If three (3) or more areas fail to meet the applicable wall thickness requirement, the sample shall be classed defective. If two or less areas fail to meet the applicable requirement, the sample shall be accepted provided that each of these areas which do not meet the requirements do not have a wall thickness of less than .080 inches for the .090 inch minimum requirement nor a wall thickness of less than .070 inches for the .080 inch minimum requirement (if applicable).

4.5.2.2 For each short divider inspected (see Dwg. AC20000402). If two (2) or more areas fail to meet the applicable wall thickness requirement, the sample shall be classed defective. If one or no areas fail to meet the applicable requirement, the sample shall be accepted provided the area which does not meet the requirement does not have a wall thickness of less than .080 inches for the .090 inch minimum requirement nor a wall thickness of less than .070 inches for the .080 inch minimum requirement (if applicable).

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4.5.3 Weld cone destructive testing (first article only). Select four long and four short dividers which have met the requirements of 4.5.2. Note that there are 30 weld cone pairs per long divider; 11 of these pairs are perpendicular to the long side, 15 are parallel, and 4 are circle cones. There are 18 weld cone pairs per short divider; 5 of these are perpendicular to the long side, 9 are parallel, and 4 are circle cones. Mark the dividers for identification of all parts prior to making the following saw cuts. Make five parallel saw cuts per two long and two short dividers so that the cuts traverse the center line of all parallel cones and cross approximately midpoint of perpendicular cones and circular cones. Make seven perpendicular cuts on the other two long dividers (five per other two short dividers) so that the cuts traverse the center line of all perpendicular weld cones and cross approximately midpoint of parallel cones and circle cones. The results of the above will allow four examinations of each weld cone pair (eight if each half is considered). All weld cones shall be found to be homogeneously welded together at the plane of the mold parting line.

4.5.4 Extreme operating temperature test (first article only). Extreme operating temperature tests will be conducted by the U.S. Army at a suitable test facility to be determined. Dividers for four racks, ammunition, 15 round assembly plus an additional 6 long and 6 short dividers (total 22 long and 30 short dividers) will be selected from the first article sample and shipped to the designated test site, (see performance test requirements specified in 3.3).

4.5.4.1 Cold test. Dividers for 2 racks plus 3 extra short and 3 extra long (15 divider, ammunition, short and 11 divider, ammunition, long) plus 2 each 15 round INERT projectile test loads (30 projectiles) are cold soaked to minus 60°F for a minimum of 24 hours. Each 15 round INERT load consist of 6 each M483, 3 each M549, and 6 each M107 projectiles, which have the same weight, center of gravity and grommets as live rounds (see 4.5.1.1).

4.5.4.1.1 Cold bend test. Two dividers, ammunition, short and two dividers, ammunition, long, are selected from the 24-hour cold soaked dividers (note, there is one extra long and short divider for use as required). Each of four dividers is to be bent 45° with the bend occurring at an assembly slot. This is accomplished by standing on one-half of the divider, reaching down and bending one end up. A second observer is to note the angle of bend. The test is to be done as soon as possible, or within the cold chamber, to avoid warming of the plastic. These dividers are not to be used for further testing, (see performance test requirements specified in 3.3).

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4.5.4.1.2 Cold vibration table test. Six short dividers and four long dividers are removed from the presoak cold chamber and assembled into a 15-projectile rack and positioned on a low frequency vibration table used for surface transportability testing. The vibration table is located within a cold chamber, and on completion of loading, the load will be reconditioned to minus 60°F. and maintained at this temperature throughout the test cycle. The rack is loaded with the cold soaked 15-projectile INERT test load and secured to the vibration table by a 1-3/4 inch nylon cargo strap in a similar manner to that used in a 5-ton tactical truck in securing the load with a long side of the rack against a side wall and with the cargo strap on three sides of the rack (see 4.5.1 and 6.6). The 15 round divider rack should be centered between two tiedown anchor points that are 54 inches apart. The proper tension on the cargo tiedown strap is achieved by first tightening to remove all slack and then adding 4 or 5 clicks more on the take-up device. The test will be run at 5 hertz for a total of 30 minutes (loose ammunition criteria). The test run includes two orientations of the rack. The first orientation is with the long side of the rack parallel to the motion of the vibration table and is to run for twenty (20) minutes. The second orientation is with the long side of the rack perpendicular to the motion of the vibration table and is to run for ten (10) minutes. The test load is to be reconditioned to minus 60°F prior to resumption of testing. This test is then repeated for the second cold rack (see performance test requirements specified in 3.3.)

4.5.4.2 Hot tests. The hot tests are the same as the cold tests except that the two racks and two INERT test loads are heat soaked to plus 145° for a minimum of 24 hours. The hot tests use the rest of the dividers which were selected at 4.5.4.

4.5.4.2.1 Hot bend test. Procedure and test criteria are the same as 4.5.4.1.1 except for the 145° temperature.

4.5.4.2.2 Hot vibration table test. Procedure and test criteria are the same as 4.5.4.1.2 except for the 145° temperature.

4.5.5 Ease of assembly and disassembly. This test shall be performed in accordance with AMC Dwg. AC200000405. In conducting this test, each divider comprising the test sample shall be used at least once. Because of variation between test sample and multiples of 6 short and 4 long dividers, some dividers may need to be used twice. In those situations where only one divider is in production, dividers from an accepted lot of the alternative length may be held back from delivery and used to complete this test.

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4.5.6 Flatness and stacking. The test sample shall be laid on a test surface which is flat within 1/32 inch and measures 44 x 26-1/2 inches. (These dimensions were selected so that stacking lugs pointing down will overhang the test surface). The sample shall be checked for flatness using a 24 inch long straight edge in compliance with 3.2.1. After checking flatness, dividers will be stacked in groups of 4, per 3.2.5. The overall height of the stack shall not exceed 4-3/4 inches excluding stacking lugs.

4.5.7 Fold lines. Critical areas for fold line inspection are designated on AMC Dwg. AC200000402 and AMC Dwg. AC200000403, as applicable. See note on applicable drawing for acceptance criteria.

4.5.8 Qualification of reworked or new mold cavity.

4.5.8.1 Qualification of a reworked mold cavity. To qualify a reworked mold cavity after production has started, at least one sample from the mold cavity shall be fully inspected at those area(s) of the part which were affected by the rework and found to meet the requirements of the basic drawing and specification. If the results of the rework are hidden from view, ie., parting line seam weld closure or weld cone weld-off etc., the part shall be cut through to determine that the mold is producing a product equal to the first article sample as tested. Established in-process quality controls which have insured an acceptable part prior to rework may be used for this inspection but must be backed up by the same examination after a curing time of 24 hours has elapsed.

4.5.8.2 Qualification of a new mold cavity. To qualify a new mold cavity after production has started will require a new First Article Inspection for that mold. The Sample shall consist of 72 Divider, Ammunition, Short if the short mold is affected or 56 Divider, Ammunition, Long if the long mold is affected. All provisions of 4.3 relating to the divider in question apply. The government will arrange for the conducting of the Performance Tests, 4.5.1 and the Extreme Operating Termperature Tests, 4.5.4 utilizing the First Article Samples and the required alternate dividers from an acceptable lot from the current production or from a previous production.

5 PACKAGING

5.1 Packing.

5.1.1 Commercial packed. Rack, 15 round assemblies are packed as a complete unit per AMC Dwg. AC200000404 sheets 1 and 2. TM 9-2590-210-10 to be inserted into each pack are government furnished. Rack assembly packs are to be arranged on a special low profile skid base for shipment as shown on AMC Dwg. AC200000404 sheet 3. The low profile skid base is detailed

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on sheet 4. Individual dividers when ordered as a single item are arranged on a pallet as shown on AMC Dwg. AC200000404 sheets 5 and 6. Individual long and short dividers will not be mixed on a pallet.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The intent of the Loose Projectile Restraint System (LPRS) is to provide reuseable dunnage to contain 155mm projectiles once they have been removed from their shipping pallets at the Field Artillery Firing Battery Area during peacetime training exercises and in wartime. It is expected that throughout its life cycle the racks will be subjected to direct or indirect natural and induced environment.

6.2 Definitions.

6.2.1 Loose Projectile Restraint System (LPRS). Refers to one or more 9-projectile, 15-projectile, or 25-projectile plastic divider racks restrained in a military tactical cargo vehicle by use of 1-3/4 inch nylon cargo tiedown straps. 155mm projectiles are placed vertically in these racks with the base of the projectile resting on the floor of the vehicle. Tiedown procedures have been developed, tested and approved by the U.S. Army Defense Ammunition Center and school (USADACS).

6.2.2 Divider. Refers to a blow molded short divider or long divider which is used to make up the three plastic divider racks. The short divider (NSN 2590-01-223-2945) is detailed on AMC Dwg. -AC200000402. The weight of the short divider is 4.2 pounds approximately. The long divider (NSN 2590-01-223-2944) is detailed on AMC Dwg. -AC200000403. The weight of the long divider is 6.4 pounds approximately.

6.2.3 Plastic divider rack or rack. Refers to a 15-projectile plastic divider rack assembly (NSN 2590-01-223-2949) illustrated on AMC Dwg. -AC200000401. (Using individual dividers, a 9 projectile rack is assembled from 8 short dividers and a 25 projectile rack from 12 long dividers.)

6.3 Ordering data. Acquisition documents should specify the following:

6.3.1 Acquisition requirements.

a. Title, number and date of this and the item detail specifications.

b. Quantity required and delivery schedules.

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c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First article sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificates of conformance for each lot or shipment of product (see 4.1.2).

6.3.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.	Options
4.4.3	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-1714	---

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Standard measuring and test equipment. Where standard measuring and test equipment (SMTE) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection and test equipment approved by the Government. The contractor shall be responsible for utilizing SMTE which will assure full form for the entire surface of the specified characteristic.

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6.5 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the Equipment Lists referenced on the Equipment Tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. However, the contractor need not furnish any design when a complete Government SIE design is part of the Technical Data Package (TDP). Unless otherwise specified, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.5.2 and 6.5.3 should he elect to do so.

6.5.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.5.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.5.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and data to which the specific design applies.

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6.5.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs, as required, to Commander, U. S. Army Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, Data Item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the Government for approval he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, mode) number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

6.5.4 Automatic inspection equipment.

6.5.4.1 Approval of automatic inspection equipment. Approval of automatic inspection equipment shall occur in two phases. The first phase will be the design phase. In this phase, designs, either conceptual or detailed, will be submitted for evaluation to the cognizant technical agency (CTA) indicated in the item specification. Design approval will authorize the contractor to proceed with fabrication. The second phase will be the equipment evaluation phase. During this phase an acceptance test will be conducted. Final approval of the inspection system will be contingent upon the performance of a successful test. A plan determining the requirements of an acceptable test will be prepared by the contractor and submitted for approval to the CTA. The test will be witnessed by DCAS, or if the equipment is of such complexity that DCAS does not feel qualified to perform the evaluation, the CTA will provide assistance.

6.5.4.2 Periodic testing. In addition to the initial acceptance test conducted in phase 2, periodic test shall be performed to verify continued system reliability. A plan will be provided by the contractor which will define the requirements for acceptable verification tests and also prescribe the frequency at which the tests will be performed. This plan will require approval of the CTA. DCAS will witness the tests or if they do not feel qualified, assistance will be provided by the CTA.

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6.6 Tie down procedures (see 4.5.1). The tie down procedures and videotape(s) of previous testing can be obtained from Director, U. S. Army Defense Ammunition Center and School, SMCAC-DEV, Savanna, IL 61074-9639 or phone (815) 273 8929, Mr. Jerry Krohn or Mr. John Sprague.

6.7 Fold lines. Fold lines occur when the plastic is not uniformly distributed over the interior surface of the mold during forming, i.e., a fold occurs in the hot plastic bag. Since there is no positive mechanical welding of the plastic along the fold, the fold line is of an unknown structural value. Additionally, the fold line robs the plastic available for distribution to the extremities of the mold. This can result in thin soft spots in the divider side walls to the extent of being translucent when sectioned and viewed from the inside.

6.8 Original specification. The original revision of this specificaton was for AMCCOM use only and not forwarded for distribution to the Navy Publications and Forms Center, Philadelphia, PA.

6.9 Subject term (key word) listing.

Projectiles
Restraint
Vehicle storage
Vehicle transport

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
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Preparing activity
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(Project 2590-A011)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-48664A	LOOSE PROJECTILE RESTRAINT SYSTEM (LPRS). 155MM

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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